Project Workflow: Week 5

Phase: Button Grid Implementation

1. Objective

Implement the calculator button grid for numeric, operator, and scientific controls, arrange the buttons using GridLayout, and apply consistent button styling.

2. Workflow

1. Add numeric buttons
2. Add operator buttons
3. Add scientific function buttons
4. Arrange the buttons using GridLayout
5. Apply button styling and mouse interaction

1. Add Numeric and Operator Buttons

- Create buttons for digits 0–9.
- Add arithmetic operator buttons: +, −, ×, and ÷.
- Add the decimal point and equals (=) button.
- Add basic control buttons such as AC, CE, and Backspace.

2. Add Scientific and Special Buttons

- Add scientific buttons such as sin, cos, tan, log, and ln.
- Add advanced buttons such as x^2, √, x^y, 1/x, and %.
- Add π, e, ±, ANS, and other required interface buttons.
- Prepare the complete button set for later calculator logic implementation.

3. Arrange Buttons Using GridLayout

- Use Java Swing GridLayout to create an organized button grid.
- Place buttons in rows and columns according to the calculator layout.

- Maintain equal spacing between buttons for a clean interface.

4. Apply Button Styling

- Apply consistent fonts, foreground colors, and background colors.
- Remove unnecessary focus and border effects for a clean appearance.
- Add mouse hover behavior to improve user interaction.
- Keep the design consistent with the calculator theme.

Week 5 Deliverables

- Numeric buttons implemented
- Operator buttons implemented
- Scientific and special buttons added
- Button grid arranged using GridLayout
- Button styling applied
- Complete calculator button interface prepared